An Inevitable Need For 5G CONVERGENCE To Serve The Massive Thrust Of DATA

Unlike previous generations of mobile networks, the fifth-generation (5G) technology is expected to fundamentally transform the role that telecommunications technology plays in the society

Saurabh Verma is founder-director and chief technology consultant, Fundarc Communication (xgnlab)

Despite continuous improvements in data services with high data rates and more spectrum utilisation, networks are still not free from congestion. The tremendous growth of video traffic and smart devices has resulted in heavy pressure on data network resources. As a result of this continuous thrust for data capacity to address data demands from specific applications, like augmented reality/virtual reality (AR/VR), massive IoT (mIoT), vehicle-to-vehicle (V2V) or vehicle-to-other-things (V2X), healthcare, remote robotics, machine-to-machine (M2M) communication and so on, cellular telecom standards generation body 3rd Generation Partnership Project (3GPP) has come out with its recent release of cellular wireless standards with the idea of incorporating more spectrum bands, including unlicensed and shared spectra.

- mIoT - massive Internet of Things
- eMBB - enhanced Mobile Broadband
- URLLC - Ultra-low latency communication

In this era of next-generation telecom networks, LTE was the first technology to step in. Being completely data-centric, it started an era of data-dependent information and communication (ICT) systems. Data has now become a utility for end-users to run their day-to-day businesses. Started 10 times faster than its predecessors, LTE is now 100 times faster in the form of gigabit LTE.

Wi-Fi, in its new avatar called 802.11ax, has become a partner for LTE, for better convergence. It provides a combination of licensed and unlicensed bands for a larger capacity to end-users.

This also opens the door for Wi-Fi's entry to 5G systems as an associative data network, with LTE as a control plane. Hence, it will not just be New Radio (NR) with LTE but Wi-Fi, too—mostly in non-standalone (NSA) mode of 5G.